K-FNSPID: Leakage-Aware Korean News and Disclosures for Target-Aware Financial Sentiment and Market-Impact Evaluation

Anonymous ACL submission

Abstract

This study introduces K-FNSPID, which links 524,696 Korean news articles and 722,989 regulatory disclosures to 10,691,998 file-based daily price records. The frozen release contains 1,247,685 documents, 1,136,118 document–security links, 715,015 market-impact rows, and 255,168 representatives after event-level deconfounding. We further specify Hana Montana AI(KF-DeBERTa + K-FNSPID), a target-aware sentiment classifier. Following an audit that rejected legacy labels without execution receipts, we re-annotated five complete packets in fresh prediction- and market-blind contexts. The resulting receipt-bound resources contain 596 main-news, 598 auxiliary-news, 600 auxiliary-disclosure, 448 development-news, and 447 development-disclosure Gold items; eleven scientifically unresolved items are excluded rather than force-labeled. The experiment separates group-disjoint checkpoint, calibration, and selection roles and prespecifies source-specific comparisons with Holm correction. In the locked 600-item-per-source confirmatory evaluation, design-weighted accuracy/macro-F1 is 0.7503/0.5530 for NEWS and 0.8646/0.6024 for DISCLOSURE. Joint source-wise superiority is not established and the deployment decision is `KEEP_CURRENT_MODEL`; therefore, this manuscript makes neither a global state-of-the-art claim nor an out-of-time generalization claim.

1 Introduction

Financial language systems are often evaluated on sequence-level sentiment or question answering, yet production monitoring asks a target-specific question: which event affects which listed security, and in what direction? A valid answer requires text, entity links, trading-session timestamps, and role-separated evaluation. We operationally define *temporal-boundary leakage* as any influence of post-boundary documents, duplicate events, or future-derived statistics on preprocessing, calibration, or model selection for an earlier partition. The term delimits the failure modes audited in this study; we do not present it as a universal taxonomy. FNSPID connects large-scale news and prices [Dong et al., 2024], whereas Korean benchmarks emphasize financial knowledge and QA [Son et al., 2024, 2025]. A Korean resource combining news, disclosures, target-aware sentiment, all-market daily prices, and explicit leakage controls remains limited.

We present K-FNSPID, a file-distributed dataset and evaluation suite for Korean financial NLP. It combines Naver-search news, OpenDART disclosures, and daily prices for listed Korean securities. Korean trading sessions, disclosure identifiers, market segments, security codes, and company aliases require market-specific normalization. The resource supports three separated tasks: target-aware sentiment, semantic materiality, and ex-post ordinal market impact. The public system using these components is named Hana Montana AI(KF-DeBERTa + K-FNSPID). Observed price movement is never used as a synonym for semantic importance or causal impact.

Our contributions are:

- a 1,247,685-document Korean news–disclosure corpus with 1,136,118 security links and 10.69M daily price rows distributed as immutable Parquet files rather than a private database;
- 1,794 receipt-bound training Gold labels, 895 development Gold labels, 32,700 Silver items, and a fail-closed dual-review protocol that excludes unresolved cases and verifies packet, run, context, and decision lineage;
- a target-aware classifier that combines leakage-purged K-FNSPID domain-adaptive pretraining, all-layer LoRA, and shared

- plus source-residual hierarchical heads with role-separated development partitions and a schedule-matched full-fine-tuning baseline; and

- a precommitted, one-shot confirmatory protocol with six Holm-corrected hypotheses, design-based finite-population inference, immutable Git attestation, and explicit limits on temporal and SOTA claims.

2 Related Work

Financial datasets and Korean evaluation. FNSPID contains 15.7M English news items and 29.7M price records for 4,775 S&P 500 firms over 1999–2023 [Dong et al., 2024]. Its document–firm–price schema informs dataset-level comparison, while its language and market scope preclude direct score comparison. KRX Bench contains 1,002 finance questions generated and filtered for Korean, US, and Japanese firms [Son et al., 2024]. FINKRX provides Korean financial instruction data and benchmark practice [Son et al., 2025]. Both evaluate knowledge or QA rather than target-aware document sentiment. MultiFinBen extends multilingual and multimodal financial evaluation [Peng et al., 2026] and is relevant to future modality expansion, not a same-task baseline.

Target-aware financial sentiment. KorFinASC provides 12,613 human-annotated Korean finance samples for aspect-level sentiment classification and evaluates target masking against nonstationary entity knowledge [Son et al., 2023]. It is the closest public Korean target-aware resource we identified, but it does not include OpenDART disclosures or K-FNSPID's source-specific, security-linked confirmatory sample; cross-dataset scores are therefore not treated as a leaderboard comparison. FinEntity annotates entity spans and three-way sentiment in financial news, demonstrating why sequence-level polarity is insufficient when one text mentions multiple entities [Tang et al., 2023]. EFSA jointly represents company, industry, event category, and sentiment at the event level [Chen et al., 2024]. These works motivate target-conditioned inputs and source-specific reporting; their languages, annotation units, and test sets differ from K-FNSPID, so their scores are not used as direct baselines. KR-FinBERT-SC is a Korean financial sentiment classifier trained after domain adaptation on news and analyst reports [Kim and Shin, 2022]. It is used both as a frozen reference and as a full-fine-tuning architecture under matched data and update schedules.

Representation, regularization, and imbalance. KF-DeBERTa is pretrained on Korean financial corpora [Jeon et al., 2023]. LoRA adapts a pretrained model through low-rank parameter updates [Hu et al., 2022]. R-Drop regularizes two dropout-induced submodels through bidirectional KL consistency [Liang et al., 2021]. Effective-number weighting models diminishing marginal information under class imbalance [Cui et al., 2019]. These methods supply the encoder, parameterization, consistency term, and weighting rationale evaluated in the proposed classifier.

Temporal shift and statistical control. Financial sentiment performance degrades under temporal distribution shift [Guo et al., 2023], motivating chronological diagnostics and explicit temporal limitations. Holm's sequentially rejective procedure controls family-wise error across prespecified hypotheses [Holm, 1979]. Survey-sampling theory supplies finite-population correction for stratified sampling without replacement [Cochran, 1977]. These results are used to specify the evaluation design before confirmatory labels are opened. Data-statement and datasheet principles motivate provenance, coverage, bias, and version records [Bender and Friedman, 2018; Gebru et al., 2021].

3 Dataset Construction

3.1 Sources and schema

Table 1 summarizes the frozen release. News is collected through Naver Search queries anchored on listed-company aliases. Disclosures originate from OpenDART. Daily adjusted prices cover 10,691,998 security-day observations and are stored in `prices_daily.parquet`; no database table is required for training or reproduction. Each document retains publication time, source type, provider, URL-derived provenance, and normalized text. Document–security relations include the mapped code and primary-link status. Market-impact rows store the effective trading date, future response features, ordinal label, split, collision flags, and event-cluster identity.

Component	Count
News documents	524,696
OpenDART disclosures	722,989
All documents	1,247,685
Document–security links	1,136,118
Market-impact rows	715,015
Unconfounded representatives	255,168
Daily price rows	10,691,998
Linked disclosure full texts	8,972
Main sentiment Gold	596
News auxiliary sentiment Gold	598
Disclosure auxiliary sentiment Gold	600
Sentiment Silver	32,700
Sentiment development Gold	895
Prepared sentiment training rows	32,907

Table 1: Frozen K-FNSPID scale. Counts are read from the release manifest and evaluation reports.

3.2 Disclosure full text and weak supervision

We joined the complete paginated OpenDART list to source identifiers and applied response-size limits, timeout and retry policies, ZIP validation, and status-code accounting. This yielded 8,972 release-linked full texts, or 1.24% of the 722,989 released disclosures. The broader internal training snapshot contains 8,976 texts because four rows do not map to release URLs. Of 5,082 final collection targets, 4,976 were stored; the 106 failures comprise 51 short contents, one DART status 013, 37 status 014 responses, and 17 public-viewer metadata misses.

Recovered text is *not* Gold. Explicit schema fields mark weak supervision and unreviewed labels. After removing Gold URLs, 8,302 real disclosures are used only as supervised candidates for semantic materiality; they are not sentiment ground truth. This separation prevents collection rules from masquerading as independent annotation.

3.3 Receipt-bound sentiment Gold and Silver

The 2026-07-16 audit found no verifiable reviewer-level execution receipt for the historical AI labels and therefore retained them only as `LEGACY_UNVERIFIED`. We did not wrap those decisions in retrospective receipts. Instead, two reviewers independently re-annotated every source packet in fresh run and context identifiers using only the immutable packet, codebook, and prompt. A third context adjudicated disagreements without access to reviewer identities or decisions. Each receipt binds the exact item set and order, codebook and prompt hashes, model version, run and context identifiers, blindness commitments, and decision bytes.

The five re-reviews comprised 600 main-news, 600 auxiliary-news, 600 auxiliary-disclosure, 450 development-news, and 450 development-disclosure candidates. Initial agreement counts were 574, 571, 554, 434, and 444; adjudication resolved 22, 27, 46, 14, and 3 disagreements. We excluded 4, 2, 0, 2, and 3 unresolved items, yielding 596, 598, 600, 448, and 447 Gold labels. Deterministic preparation produces 32,907 training rows and group-disjoint development roles of 911 Checkpoint, 455 Calibration, and 461 Selection rows. All 15 pairwise provenance-group overlaps among training, development, and the two 600-item confirmatory reservations are zero.

Semantic-materiality evaluation remains separate: 600 primary disclosures and a 310-case stress extension follow a four-level codebook. Sentiment Gold, semantic-materiality Gold, and weakly supervised full-text disclosures have different labels and roles and are not pooled into one performance denominator.

3.4 Trading-session normalization

For a document released before the market close, the effective date is the same valid trading day; releases after close, on weekends, or on holidays map to the next trading day. Exchange calendars are applied before response computation. We cluster repeated reporting of an event using a fingerprint that includes effective date and normalized content. If multiple distinct event clusters occur for the same security and trading day, all such rows are flagged as confounded and excluded from the primary training set. Within the same event/security/day cluster, the most informative document becomes the representative.

4 Tasks and Labels

Sentiment. Given a source document and target security, the task predicts negative, neutral, or positive sentiment toward that target. NFKC, casefold, and alphanumeric normalization remove exact normalized duplicates and label conflicts. The receipt-bound 895-item development Gold and 932 development-eligible public Validation items are assigned by a fixed group key to mutually exclusive Checkpoint ($n = 911$), Calibration ($n = 455$), and Selection ($n = 461$) roles after duplicate and protected-group removal. These roles are adaptive development evidence, not confirmatory evidence. The historical public Test is excluded from all three roles and retained only for diagnostics.

Semantic materiality. The four-level codebook estimates the business significance of a disclosure from its meaning. This label must remain stable even when observed market response is small.

Ex-post market impact. Let P be adjusted close and r_m the same-market mean return. For horizons $h \in \{1, 3, 5\}$ we subtract compounded market return from compounded security return. The score applies weights of 0.50, 0.20, 0.15, and 0.15 to clipped one-day absolute excess return, three-day absolute excess return, a 60-day log-volume z-score shock, and a 20-day intraday-volatility shock, respectively, and clips the sum to $[0, 1]$. We assign LOW for $0 \leq s < 0.20$, MEDIUM for $0.20 \leq s < 0.45$, HIGH for $0.45 \leq s < 0.75$, and CRITICAL for $0.75 \leq s \leq 1$. Because future observations are used only to construct targets, the label is a weak ordinal proxy for observed response rather than causal materiality or a tradable signal.

5 Leakage-Aware Experimental Protocol

5.1 Chronological split

The training harness materializes source-specific representative sets. News contains 99,826 training, 6,391 validation, and 9,560 test rows; disclosure contains 119,146, 584, and 4,615 respectively. Validation begins 2026-01-01 and test begins 2026-04-01. Seven days immediately before each boundary are embargoed. Preprocessing, prior and temperature correction, model selection, and early stopping use no test observations. Paired comparisons require identical document IDs and labels within each source.

5.2 Models

Sentiment. We begin from `kakaobank/kf-deberta-base` at immutable revision `363b171d...0633`. Before supervised training, sparse masked-language-model adaptation uses 1,118,291 temporally eligible and protected-component-purged K-FNSPID documents packed into 62,468,526 non-padding tokens. The DAPT stage freezes the base, embeddings, and MLM head and trains rank-16 LoRA on query and value projections in all 12 layers for 3,908 updates. A paired 1,024-pack precision pilot selects FP32 because BF16 fails the prespecified MPS memory-slope gate. The resulting adapter is safely merged into an FP32 base only after input, pack, pilot, oracle, and validation-NLL commitments pass.

Supervised adaptation applies rank-16 LoRA to query, key, value, and dense projections in all 12 layers. The input prepends the target security and retains the document head and tail within 256 tokens. A shared hierarchical head first predicts neutral versus directional sentiment and then negative versus positive direction; NEWS, DISCLOSURE, and PUBLIC residual heads are initialized to exact zero, so the initial model equals the shared head. Stage 1 trains encoder and heads for two epochs under fixed domain–task-cell weight mass, whereas Stage 2 freezes the encoder and refines the heads for four epochs using receipt-bound Gold only. R-Drop supplies bidirectional KL consistency between dropout passes [Liang et al., 2021]. Target-swap augmentation is accepted only when the official company name, security code, and every registered alias are absent from the swapped context.

Data selection uses seed 20260715. Model stochasticity is assessed with seeds 17, 42, and 73; all runs must reproduce identical data and partition commitments. Checkpoint rows select training checkpoints, Calibration rows fit logit correction, and Selection rows choose the seed and final candidate. No role may be reused for another decision.

The fair comparator full-fine-tunes all parameters of `snunlp/KR-FinBert-SC`. It receives the same source rows, group partitions, data seed, model seeds, epochs, and optimizer-update schedule. Architecture-specific learning rates and trainable parameter counts are reported rather than described as identical regularization. Frozen KR-FinBERT-SC inference and the pre-K-FNSPID service model are retained as named diagnostic comparators.

Semantic materiality. We compare title, title+summary, and title+summary+full-text TF-IDF logistic models using only chronological validation. Title+summary and title tie on validation macro-F1 (0.9894); title+summary wins on Brier score (0.00134 versus 0.00156). Full text reaches 0.9815 macro-F1 and is not forced into deployment. A narrow deterministic floor covers existential-risk phrases. We report model-only and policy-augmented results separately.

Market impact. We train separate source experts because news and disclosures differ in style, temporal density, and label prevalence. Each source has a train-only class-balanced one-vs-rest logistic baseline over character 2–5-gram TF-IDF features. The neural experts use the immutable KF-DeBERTa revision with rank-16 LoRA. The disclosure expert uses 128 tokens, effective batch size 32, learning rate 5×10^{-5}, focal loss ($\gamma = 1.0$), ordinal CDF weight 0.20, and label smoothing 0.01. Validation jointly selects a log-prior offset and temperature. Disclosure runs seeds 17, 42, and 73 under the same configuration; seed 17 is selected only by validation macro-F1. The news expert uses the previously validated adapter, is frozen before the v4 test is read, and is re-evaluated on the isolated news split. Serving rejects a request when its source does not match the artifact metadata.

5.3 Metrics and inference

For sentiment, accuracy and macro-F1 are reported separately for NEWS and DISCLOSURE. The canonical statistical analysis plan (SAP) fixes six primary comparisons: the new candidate against frozen KR-FinBERT-SC, the pre-K-FNSPID model, and the fair full-fine-tuned KR-FinBERT-SC, in each source. The family-wise level is 0.05 and Holm correction is applied across all six p-values [Holm, 1979].

Confirmatory items are sampled by source and sentiment stratum using simple random sampling without replacement (SRSWOR). Paired estimands use the corresponding stratified design weights. Primary variance estimation uses a paired stratified delete-one jackknife with finite-population correction (FPC); a 2,000-replicate paired bootstrap with fixed seed 20260715 is retained for interval and sensitivity analysis. Weighted macro-F1 is a nonlinear plug-in estimator and is not described as exactly unbiased. Holm correction controls the p-value family; it does not turn pointwise confidence intervals into simultaneous intervals.

The public sentiment Test was repeatedly inspected in earlier development and is diagnostic only. Model, recipe, partition commitments, and the SAP must be attested by a pushed Git commit before confirmatory labels are completed. The confirmatory evaluator runs once and emits a non-reusable receipt. Market-impact evaluation remains a separate protocol reporting accuracy, macro-F1, quadratic weighted Cohen's kappa (QWK), ordinal mean absolute error, 15-bin expected calibration error (ECE), and multiclass Brier score [Guo et al., 2017]; its trading-day cluster bootstrap is not reused as the sentiment sampling design.

6 Results

6.1 Sentiment

Table 2 reports the receipt-bound one-shot evaluation produced after the pushed lock commit and post-lock Gold completion. Historical public-Test scores remain engineering diagnostics and are excluded. The primary estimands are sampling-design-weighted accuracy and macro-F1; no improvement, statistical superiority, deployment promotion, or state-of-the-art claim is made beyond the locked source-specific results.

6.2 Disclosure semantic materiality

The title+summary model alone reaches 0.9850 accuracy, 0.9470 macro-F1, and 0.8163 CRITICAL F1 on the 600-case primary Gold, compared with 0.7150/0.4489/0.2619 for the pre-v3 integrated analyzer. Adding the narrow codebook floor yields 1.0000 on the primary set. On primary plus stress Gold (910 cases), the operational candidate reaches 0.9989 accuracy, 0.9962 macro-F1, 0.9994 QWK, 0.0225 ECE, and 0.0063 Brier. The exact McNemar p against the previous analyzer is 1.14×10^{-24}. These values measure agreement with the same codebook used to design the floor; they cannot establish independent expert validity. Model-only performance is therefore the less circular primary scientific result.

6.3 Temporally extrapolated market impact

Table 3 reports the frozen source-specific tests. The news expert improves macro-F1 from 0.3484 to 0.3745 and QWK from 0.3421 to 0.4754. The selected disclosure expert improves macro-F1 from 0.2677 to 0.3216 and QWK from 0.1125 to 0.1550. Its three-seed test macro-F1 is 0.3170 ± 0.0052; seed 17 is selected by validation macro-F1 0.2471, without consulting test. Combined source-routed predictions improve accuracy from 0.4377 to 0.5085, macro-F1 from 0.3210 to 0.3690, and

Model/comparison	NEWS	DISCLOSURE
Hana Montana AI(KF-DeBERTa + K-FNSPID)	0.7503 / 0.5530	0.8646 / 0.6024
Raw KR-FinBERT-SC	0.5781 / 0.4937	0.8535 / 0.6146
Pre-K-FNSPID model	0.4737 / 0.4396	0.8480 / 0.5358
Fair full-FT KR-FinBERT-SC	0.7677 / 0.5771	0.8514 / 0.5647
Holm-adjusted decision	Not superior	Not superior

Table 2: Locked confirmatory sentiment results on 600 NEWS and 600 DISCLOSURE items. Each numeric cell reports sampling-design-weighted accuracy/macro-F1. The candidate was not promoted.

Source/model	Acc.	Macro-F1	QWK
News TF-IDF	0.4715	0.3484	0.3421
News KF-DeBERTa	**0.5247**	**0.3745**	**0.4754**
Disc. TF-IDF	0.3675	0.2677	0.1125
Disc. KF-DeBERTa†	**0.4750**	**0.3216**	**0.1550**
Combined routed	0.5085	0.3690	0.3975

Table 3: Market-impact test results (news $n = 9{,}560$, disclosure $n = 4{,}615$). † Disclosure seed selected only by validation macro-F1.

Source	Model	n	M-F1	QWK
News	TF-IDF	9,560	0.3484	0.3421
	KF-DeBERTa		**0.3745**	**0.4754**
Disclosure	TF-IDF	4,615	0.2677	0.1125
	KF-DeBERTa		**0.3216**	**0.1550**

Table 4: Source-stratified market-impact test after source-routed training and calibration.

QWK from 0.2552 to 0.3975. ECE also improves from 0.0369 to 0.0259.

6.4 Paired statistical inference

Trading-day cluster-bootstrap 95% CIs for Transformer-minus-baseline macro-F1 are [0.0120, 0.0409] for news and [0.0264, 0.0806] for disclosure; QWK intervals are [0.1090, 0.1558] and [0.0046, 0.0794]. Combined intervals are accuracy [0.0594, 0.0818], macro-F1 [0.0351, 0.0608], and QWK [0.1178, 0.1649]. Exact McNemar $p = 2.29 \times 10^{-55}$. Thus, the earlier disclosure regression is removed without hiding source results. The lower disclosure QWK bound remains close to zero, so replication across additional periods is necessary.

7 Ablations and Operational Comparison

The previous shared model had only 590 disclosure test rows and regressed from 0.3006 to 0.2211 macro-F1. Expanding disclosure metadata to 722,989 documents yields 4,615 deconfounded disclosure test rows; source-specific fine-tuning lifts macro-F1 to 0.3216. Historical TF-IDF ablations also showed title+snippet outperforming full text (0.3505 versus 0.3429 on the former mixed split), so we preserve 8,972 linked disclosure full texts without claiming that context length alone caused the gain. The source route, larger disclosure split, and validation-only calibration change together; a factorial attribution remains future work.

The sentiment redesign changes several factors jointly: receipt-bound labels, DAPT, target-aware input, all-layer LoRA, shared and source-residual heads, R-Drop, fixed cell-mass weighting, alias-safe counterfactuals, and three-way development roles. Confirmatory Gold cannot be used for ablation or attribution. Accordingly, component ablations are restricted to the receipt-bound development Selection split and are reported as exploratory; only the fully locked recipe enters one-shot confirmation. No performance gain is attributed to any single component before the development ablations and confirmatory receipt are available.

The FULL reference arm reuses the locked candidate artifact from the same seed instead of retraining it. A content-addressed receipt revalidates the candidate report, input and partition commitments, DAPT base, update budgets, CPU round trip, and every artifact-manifest byte count and SHA-256. The candidate-native minibatch sequence is retained; we therefore claim matched source rows, exposure counts, and optimizer-update budgets, not bitwise identity with the stable-order removal arms.

8 Discussion

K-FNSPID advances Korean financial NLP primarily as infrastructure and evaluation discipline, not as a solved prediction task. Its strongest empirical finding is that one shared market-impact model can conceal a disclosure regression, while source-routed experts improve both sources under chronological evaluation. The second finding is methodological: semantic importance and price impact must be separate API outputs. A bankruptcy disclosure can be intrinsically critical even if anticipated by the market; a rumor can cause a large move without being semantically reliable.

Compared with FNSPID, the corpus is smaller and covers a different language and market. Compared with KRX Bench and FINKRX, it evaluates event-linked classification rather than QA. Accordingly, we claim neither cross-benchmark SOTA nor profitable forecasting. For sentiment, the candidate exceeds raw KR-FinBERT-SC macro-F1 on NEWS by 0.0594 but trails it on DISCLOSURE by 0.0123; Holm-adjusted joint superiority is not established. Market-impact claims remain limited to the already frozen source-matched tests.

9 Conclusion

We introduced K-FNSPID, a leakage-aware Korean news–disclosure–price resource, and specified Hana Montana AI(KF-DeBERTa + K-FNSPID), its target-aware sentiment and materiality stack. The release connects 1.25M documents to 10.69M daily price rows. The sentiment protocol contributes a fail-closed dual-review receipt chain, matched full-fine-tuning baselines, disjoint development roles, and a locked design-based confirmatory analysis. The one-shot evaluation yields 0.7503/0.5530 weighted accuracy/macro-F1 on NEWS and 0.8646/0.6024 on DISCLOSURE, does not establish joint superiority, and retains the current production model. Independent human-expert annotation and a genuinely later out-of-time evaluation remain necessary.

Limitations

News collection is subject to Naver Search result limits, query design, provider availability, deduplication, and survivorship effects. OpenDART full-text coverage is 1.24%, so missingness is not random across disclosure types or dates. Company alias mapping can miss renamed firms, ambiguous abbreviations, and historical securities. Although prices cover the full stored universe, corporate actions and exchange-specific data corrections may remain.

Daily data cannot isolate intraday reaction, information leakage before publication, or response outside the chosen 1/3/5-day windows. Market-mean subtraction does not remove industry, size, macroeconomic, or concurrent-event confounding. Event clustering and security-day collision filtering reduce obvious contamination but cannot identify every related story. Therefore, the impact target is correlational and weakly supervised.

The public sentiment Test is balanced, may not match production prevalence, and was adaptively inspected during earlier development. It therefore remains diagnostic regardless of any new model score. Receipt-complete re-review establishes procedural separation among AI review contexts, but not agreement among independent human financial experts or external construct validity. The Gold resources should therefore be interpreted as reproducible codebook-conforming AI adjudication, not as a substitute for human-expert validation.

Development Gold is concentrated in June–December 2025, whereas auxiliary disclosure Gold and confirmatory candidates are concentrated in April–July 2026. The confirmatory sample is document-disjoint and committed before labeling, but it is contemporaneous with part of the training resource rather than a clearly later out-of-time period. A one-shot result can therefore support disjoint same-period confirmation, not long-horizon temporal generalization. Eleven unresolved cases are excluded across the five reviewed packets; excluding them is safer than forcing labels but can narrow coverage of truncated or ambiguously linked texts.

Fixed seeds reproduce data commitments and control model repetitions but do not guarantee bitwise-identical kernels across Apple MPS and other accelerators. Each report records the actual device, global step, and runtime; procedural repeatability is distinguished from bitwise reproducibility.

The market-impact test spans only 70 effective trading dates. Disclosure has 4,615 rows across 34 dates, and its QWK gain interval begins at 0.0046; broader periods may reverse the result. Korean-only text and the selected 2026 boundary limit geographic and temporal generalization. Base-model revision pinning improves reproducibility but can preserve unknown model biases. No result should be interpreted as financial advice, causal effect, or expected return.

Ethical Considerations

The sources are publicly accessible news, disclosures, and market records, but public availability does not eliminate copyright, privacy, or contextual-integrity concerns. Distribution follows the project's declared research-use policy and preserves provenance and deletion/version procedures. Credentials, API keys, and private user data are excluded. URLs and identifiers can still expose individuals named in corporate events; downstream users should apply access controls, retention limits, and purpose limitation.

Financial NLP can amplify rumors, market manipulation, or automation bias. The API returns semantic materiality and observed market-impact estimates separately, includes confidence and model version, and fails closed to validated fallbacks when artifact hashes or promotion gates fail. It does not execute trades, generate personalized recommendations, or claim causality. High-stakes alerts require human review and primary-source verification.

The author used generative coding assistants for data engineering, code review, independent codebook-based annotation passes, adjudication, and manuscript drafting. Numeric claims are admitted only from versioned machine-readable reports and the confirmatory claims are bound to the locked report hash. This assistance is disclosed because model-family correlation can remain even across separate review passes and does not substitute for independent domain experts.

Reproducibility Statement

The environment uses Python 3.12 with a locked dependency graph. Reports record the KF-DeBERTa revision, DAPT and supervised all-layer LoRA scopes, data seed, model seeds, optimizer, scheduler, completed epochs, and each development-role decision. Large Parquet snapshots are versioned through Git LFS, while the release manifest independently records original SHA-256 values and byte sizes. The upstream base provides PyTorch weights; the exact source hash is verified with safe weight-only loading before conversion to deployment safetensors. Remote model code is disabled, and the runtime verifies version, source, path, size, signature, and hash before activation. The training recipe, dependency lock, candidate lock, reservation commitments, and canonical SAP are content-addressed in a pushed Git attestation. Confirmatory evaluation additionally requires a one-shot receipt. The anonymous supplement includes the datasheet, codebook, report JSON, verifier, and build instructions; its public URL is withheld during review.

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A Release Files

The frozen distribution contains `documents.parquet`, `document_entities.parquet`, `market_impacts.parquet`, `prices_daily.parquet`, `splits.parquet`, and `annotations.parquet`. The corresponding byte sizes and SHA-256 values are recorded in `manifest.json`. Dataset data are never reconstructed from an application database.

B Promotion Gates

Sentiment promotion requires (i) identical input and development commitments across the three candidate and fair-baseline seeds, (ii) a candidate and canonical SAP attested by a pushed remote commit before confirmatory labeling, (iii) one-shot NEWS and DISCLOSURE evaluation with a valid receipt, (iv) source-specific absolute accuracy and macro-F1 gates, (v) non-regression against the fair full-fine-tuned KR-FinBERT-SC comparator, and (vi) signed artifact-integrity validation. The final decision is `KEEP_CURRENT_MODEL`: the candidate failed the absolute gates and did not establish source-wise Holm-adjusted superiority. The historically reused public Test cannot satisfy a promotion or confirmatory gate. Market-impact promotion remains governed by its separately frozen source-specific gates and does not borrow the sentiment confirmatory sample.